ECE 3300 Lab 1 – Switch ↔ LED Interface

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Objectives:
– Build a Verilog module on the Digilent Nexys A7-100T that reads 16 switches and drives 16 LEDs.
– Learn HDL I/O mapping, constraints files, synthesis, and FPGA programming

Report: Verilog code, XDC snippet, synthesis screenshots (LUT & FF counts), group video link, reflections

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Verilog code screenshot:

Project Summary | Device | switch_led_interface.v | Nexys-A7-100T-Master.xdc

C:/Users/ttbat/lab 1/lab 1.srcs/sources_1/new/switch_led_interface.v

```verilog
// Tool Versions:
// Description:
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
//////////////////////////////////////////////////////////////////////////////////


module switch_led_interface(
    input wire [15:0] sw,
    output wire [15:0] led
);
    assign led = sw;
endmodule
```

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XDC snippet:

##Switches
#set_property -dict { PACKAGE_PIN J15   IOSTANDARD LVCMOS33 } [get_ports { SW[0] }]; #IO_L24N_T3_RS0_15 Sch=sw[0]
#set_property -dict { PACKAGE_PIN L16   IOSTANDARD LVCMOS33 } [get_ports { SW[1] }]; #IO_L3N_T0_DQS_EMCCLK_14 Sch=sw[1]
#set_property -dict { PACKAGE_PIN M13   IOSTANDARD LVCMOS33 } [get_ports { SW[2] }]; #IO_L6N_T0_D08_VREF_14 Sch=sw[2]
#set_property -dict { PACKAGE_PIN R15   IOSTANDARD LVCMOS33 } [get_ports { SW[3] }]; #IO_L13N_T2_MRCC_14 Sch=sw[3]…

LEDs
#set_property -dict { PACKAGE_PIN H17   IOSTANDARD LVCMOS33 } [get_ports { LED[0] }]; #IO_L18P_T2_A24_15 Sch=led[0]
#set_property -dict { PACKAGE_PIN K15   IOSTANDARD LVCMOS33 } [get_ports { LED[1] }]; #IO_L24P_T3_RS1_15 Sch=led[1]
#set_property -dict { PACKAGE_PIN J13   IOSTANDARD LVCMOS33 } [get_ports { LED[2] }]; #IO_L17N_T2_A25_15 Sch=led[2]
#set_property -dict { PACKAGE_PIN N14   IOSTANDARD LVCMOS33 } [get_ports { LED[3] }]; #IO_L8P_T1_D11_14 Sch=led[3]
#set_property -dict { PACKAGE_PIN R18   IOSTANDARD LVCMOS33 } [get_ports { LED[4] }]; #IO_L7P_T1_D09_14 Sch=led[4]…

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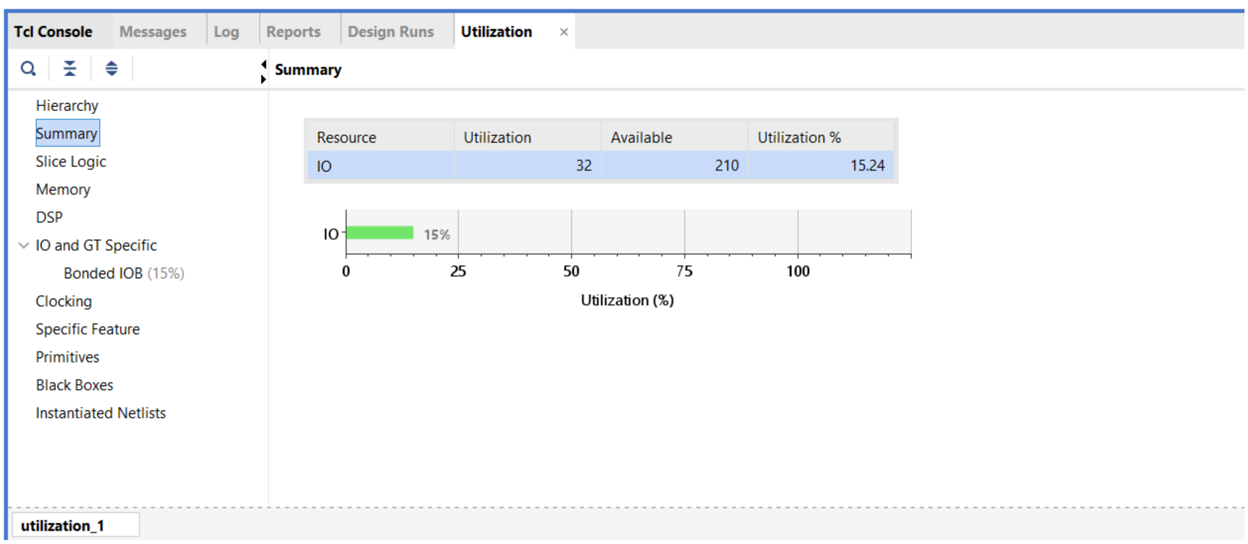
Synthesis screenshots



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Video link: https://youtu.be/vLWRK_sMb9I

Conclusion:
This lab was fairly straight forward in being a simple task to connect switches to led reactions in verilog.